Solutions
QA : Algebra - 05

1. 3 $7 + 3|x| = 13 \Rightarrow 3|x| = 6 \Rightarrow |x| = 2$
So, $x = \pm 2$

Alternative method:
Use the options and try.

2. 2 Since x^2 cannot be less than 0 for any value of x, so $|x^2+1| \le 1$ will hold only when x = 0.
Hence, $(x+2)(x+1)(x)(x-1)(x-2)\cdots = 0$

3. 4 $|3x| + 7|x| \le 60$
For $x \ge 0$
$3x + 7x \le 60$
$x \le \frac{60}{10} \Rightarrow x \le 6$
For $x < 0$
$-3x + 7(-x) \le 60$
$\Rightarrow -10x \le 60 \Rightarrow x \ge -6$
$\Rightarrow$ value of x is $-6 \le x \le 6$

Alternative method:
The inequation given in the question contains $\le$ but only option (4) contains equals to sign. So, option (4) is correct.

4. 3 To make $\frac{a^2}{b^2}$ minimum, a must be minimum and b must be maximum. Also, a and b are +ve integers, so from the given inequations
a = 1, 2, 3, 4, 5 and b = 1
So, minimum value of $\frac{a^2}{b^2} = 1$.

5. 3 Given 3x + 2|y| + y = 7 and x + |x| + 3y = 1

If x > 0, y > 0	3x + 2y + y = 7 x + x + 3y = 1	We get x = 6 and y = –ve This is a contradiction as y < 0.
If x > 0, y < 0	3x – 2y + y = 7 x + x + 3y = 1	We get x = 2 and y = –1. This is a possible solution
If x < 0, y > 0	3x + 2y + y = 7 x – x + 3y = 1	We get x = 2 and y = 1/3 This is a contradiction as x > 0.
If x < 0, y < 0	–3x – 2y + y = 7 x – x + 3y = 1	We get y = 1/3 and x = –ve This is a contradiction as y > 0.

So x + 2y = 2 - 2 = 0 for (x, y) = (2, -1).

6. 4 The given equation will have infinite number of solutions if x in |x + a| and x in |x – 1| cancel out.
Case 1: x + a < 0 and x – 1 ≥ 0
$\Rightarrow -a - x + x - 1 = 2 \Rightarrow a = -3$
Case 2: x + a = 0 and x – 1 < 0
$\Rightarrow x + a - x + 1 = 2 \Rightarrow a = 1$
Hence, the largest value of a that will give an infinite number of solutions for the given equation is 1.

For questions 7 and 8:
Let |x| = m and |y| = n
m + n = 7 and $m^2 + n^2 = 25$
or $m^2 + (7 - m)^2 = 25$
Solving above equation we get m = 3 or 4 and hence n = 4 or 3 or $x = \pm 3$ or ± 4 and $y = \pm 4$ or ± 3

7. 4 It doesn't matter what value is assigned to x and y among ± 3 and ± 4 because the expression is $x^3 + y^3$
Cleary a total of four values are possible for $x^3 + y^3$. This will happen as (x, y) is retrieved from the following sets.
(– 3, – 4) or (+ 3, + 4) or (– 3, + 4) or (3, – 4)

8. 4 Here, x and y can be +ve and –ve both.
So, value of $|x - y|^2$ cannot be determined uniquely.

9. 4 $||2x-19|-7| < 5$
$-5 < |2x-19| - 7 < 5$
$\Rightarrow 2 < |2x-19| < 12$
$\Rightarrow -12 < 2x - 19 < -2$ or $2 < 2x - 19 < 12$
$\Rightarrow 7 < 2x < 17$ or $21 < 2x < 31$
$\Rightarrow \frac{7}{2} < x < \frac{17}{2}$ or $\frac{21}{2} < x < \frac{31}{2}$
Therefore, there are 10 integer values of x that satisfy the inequality i.e. x = 4, 5, 6, 7, 8, 11, 12, 13, 14 and 15.

10. 1 |n – 60| < |n – 100| < |n – 20|
|n – 60| < |n – 100| (60 < n < 100)
Or, n – 60 < 100 – n
Or, 2n < 160
Or, n < 80
|n – 100| < |n – 20| (20 < n < 100)
Or, 100 – n < n – 20
Or, 2n > 120
Or, n > 60
Hence, number of integer values of n = 61 to 79 = 19.

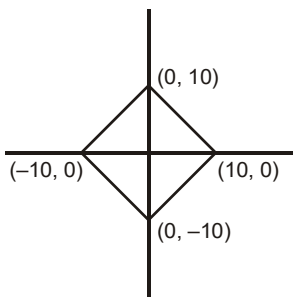
11. 4 Equation 1:
$3x - 20 + 3x - 40 = 20$
$6x - 60 = 20$
$6x = 80$
$x = 80/6 = 13.33$
Equation 2:
$-(3x - 20) - (3x - 40) = 20$
$-6x + 60 = 20$
$-6x = -40$
$x = 40/6 = 6.67$
As per the options, for $7 < x < 12$, the condition necessarily holds.

12. (1) The minimum value will occur at $x = 2$, which is 3.
(2) The minimum value will occur at $2 \le x \le 4$, which is 9.

13. 12 Given : $|1 + mn| < |m + n| < 5$
This inequality is true for (m, n) or (n, m)
$= (\pm 2, 0), (\pm 3, 0), (\pm 4, 0)$
Hence, there are 12 possible pairs of integers.

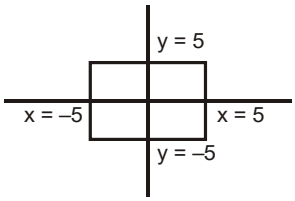
14. 2 Given $0 \le a \le x \le 100$
$f(x) = |x - a| + |x - 100| + |x - a - 50|$
Since $x \ge a \Rightarrow |x - a| = x - a$
and $x \le 100 \Rightarrow |x - 100| = 100 - x$
$\Rightarrow f(x) = x - a + 100 - x + |x - a - 50|$
$f(x) = 100 - a + |x - a - 50|$
Let us go by options.
(1) Put $a = 0$; $f(x) = 150$ if $x = 100$
150 if $x = 0$
(2) Put $a = 25$; $f(x) = 125$ if $x = 25$
(3) Put $a = 50$; $f(x) = 100$ if $x = 50$
(4) Put $a = 100$; $f(x) = 50$ if $x = 100$
we can see that max $f(x) = 100$
$f(x) = 100$ for $a = 50$.

15. (1) The graph of the function $|x| + |y| = 10$ is given below



So, area = $4 \times \frac{1}{2} \times 10 \times 10 = 200$ sq. unit
(2) The given region is same as previous one, but the origin is shifted. So, area remains unchanged, which is 200 sq. unit

(3) The graph of $|x - y| + |x + y| = 10$ is given as



So, required area = 10 × 10 = 100 sq. unit.

16. 3 Here, $a = d^3$, $b = d^3$ and $c = d^{\frac{6}{5}}$.

$\therefore \log_d(abc) = \log_d d^{3+3+\frac{6}{5}} = 7.2\log_d d = 7.2.$

17. 1 $\log_{210}5 + \log_{210}6 + \log_{210}7$
$\Rightarrow \log_{210}5 \times 6 \times 7 = \log_{210}210 = 1$

18. 4 $\log_2 \log_3 \log_2\log_x 2^{1024} = 1$
$\Rightarrow \log_3 \log_2\log_x 2^{1024} = 2^1$
$\Rightarrow \log_2\log_x 2^{1024} = 3^2 = 9$
$\Rightarrow \log_x 2^{1024} = 2^9 = 512$
$\Rightarrow 2^{1024} = x^{512} \Rightarrow 4^{512} = x^{512}$
$\Rightarrow x = 4$

19. 4 Here $3\log_{\frac{(x+4)}{8}}(x-1) = \log_{\frac{(x+4)}{8}} 27$

$\Rightarrow \log_{\frac{(x+4)}{8}}(x-1)^3 = \log_{\frac{(x+4)}{8}} 27$

So, $(x - 1)^3 = 27$
$\Rightarrow x - 1 = 3 \Rightarrow x = 4$

Here, $\frac{x+4}{8}$ cannot be equal to 1.

At $x = 4$, $\frac{x+4}{8} = 1$ --- a contradiction.
So, no solution is possible.

20. 2 $P = 27^{\log_3 48} \times 3^{\log_{27} 48^2} \times 27^{\log_3 48^3} \times 3^{\log_{27} 48^4} \times \ldots$

$\ldots \times 27^{\log_3 48^{99}}$.

$\Rightarrow P = 48^3 \times 48^{\frac{2}{3}} \times 48^9 \times 48^{\frac{4}{3}} \times \ldots\ldots \times 48^{297}$

$\Rightarrow P = 48^{3(1+3+5\ldots\ldots+99)} \times 48^{\frac{2}{3}(1+2+3+\ldots\ldots+49)}$

$\Rightarrow P = 48^{\frac{24950}{3}} \Rightarrow x = \frac{24950}{3}$

$\therefore \frac{3x}{50} = 499.$

21. 2 $81^{\log_{10} x} = 3 + 2 \times x^{\log_{10} 9}$

$9^{2\log_{10} x} = 3 + 2 \times 9^{\log_{10} x}$ $\left(\text{because } a^{\log_{10} b} = b^{\log_{10} a}\right)$

Let $9^{\log_{10} x} = k$.

$\Rightarrow k^2 = 3 + 2k$

Therefore, k = 3 or –1
But k cannot be negative.

$\Rightarrow 9^{\log_{10} x} = 3$

$\therefore x = \sqrt{10}$

Hence, x has only one value.

22. 1 $2\left[27^{\log_3(\log_4 x)}\right] = (\log_4 x)^2 + 5(\log_4 x) + 2$

Putting $\log_4 x = y$, we get

$\Rightarrow 2y^3 = y^2 + 5y + 2$

$\left[\because z^{\log_z m} = m, \text{if } z \text{ is any constant}\right]$

$\Rightarrow 2y^3 - y^2 - 5y - 2 = 0$

$\Rightarrow (y+1)(y-2)(y+\frac{1}{2}) = 0 \Rightarrow y = -1, 2, -\frac{1}{2}$.

But $\log_4 x$ cannot be negative.
So, y = 2
$\log_4 x = 2$
$\Rightarrow x = 4^2 = 16$
$\therefore$ Number of values of x is 1.

23. 2 Log 3^{-70} = (–70) Log 3 = (–70) × (0.477) = –33.39
= –34 + 0.61 = $\overline{34}.61$
So 3^{-70} should be in the form of A × 10^{-34}.
So the answer is 33.

24. 4 $\log_x\left(\frac{x}{y}\right) + \log_y\left(\frac{y}{x}\right)$

$= \log_x x - \log_x y + \log_y y - \log_y x$

$= 2 + \log_x y + \frac{1}{\log_x y}$

Since $\log_x y$ is a non-negative number, the value of $\left(\log_x y + \frac{1}{\log_x y}\right)$ has to be greater than or equal to 2.

So, the value of the given expression cannot be a positive real number. Thus, 1 cannot be the value of the given expression.

25. 4 Since $\log_2 x + \log_x 2 = \frac{10}{3} = \log_2 y + \log_y 2$ and $x \neq y$,

which means if we convert $\log_2 x + \log_x 2 = \frac{10}{3}$ into a quadratic equation, then x will take two values, same would be for y. i.e. we have to find the sum of the roots.

So, $\log_2 x + \log_x 2 = \frac{10}{3}$

$\log_2 x + \frac{1}{\log_2 x} = \frac{10}{3}$

$3(\log_2 x)^2 - 10(\log_2 x) + 3 = 0$

So, $\log_2 x + \log_2 y = \frac{-(-10)}{3} = \frac{10}{3}$.

26. 3 $\log_7 \log_5\left(\sqrt{x+5} + \sqrt{x}\right) = 0$

For the equation to be 0, $\sqrt{x+5} + \sqrt{x}$ must be equal to 5. i.e. $\log_5\left(\sqrt{x+5} + \sqrt{x}\right) = 1$.

Putting x = 4, satisfies the equation.

27. 2 $\log_{10} x - \log_{10}\sqrt{x} = 2\log_x 10$

$\log_{10}\left[\frac{x}{\sqrt{x}}\right] = \log_x 100$

$\therefore \log_{10}\sqrt{x} = \frac{\log_{10} 100}{\log_{10} x}$

$\therefore \frac{1}{2}\log_{10} x = \frac{2}{\log_{10} x}$

$\therefore (\log_{10} x)^2 = 4$

$\therefore \log_{10} x = \pm 2$

$\therefore \log_{10} x = 2$ or $\log_{10} x = -2$

$\therefore 10^2 = x$ or $10^{-2} = x$

$\therefore x = 100$ or $x = \frac{1}{100}$

28. 4 $\log_y^x = a.\log_z^y = b.\log_x^z = a \times b$

$a = \dfrac{\log_y^x}{\log_z^y}$ and $b = \dfrac{\log_y^x}{\log_x^z}$

$\Rightarrow a \times b = \dfrac{\log_y^x}{\log_z^y} \times \left(\dfrac{\log_y^x}{\log_x^z}\right)$

$= \dfrac{\left(\dfrac{\log_k^x}{\log_k^y}\right)}{\left(\dfrac{\log_k^y}{\log_k^z}\right)} \times \dfrac{\left(\dfrac{\log_k^x}{\log_k^y}\right)}{\left(\dfrac{\log_k^z}{\log_k^x}\right)}$ {For some base k}

$= \left(\dfrac{\log_k^x}{\log_k^y}\right)^3 = \left(\log_y^x\right)^3 = (ab)^3$

So, $ab - a^3b^3 = 0$

or, $a \times b(1 - a^2b^2) = 0$

$\Rightarrow ab = \pm 1$

Only option (4) does not satisfy.

29. 2 Since the given terms are in A.P.

So, $2 \times \log_4(2^{(1-x)} + 1) = \log_2(5 \times 2^x + 1) + 1$

$\log_2(2^{(1-x)} + 1) = \log_2(5 \times 2^x + 1) + \log_2 2$

$\Rightarrow 2^{1-x} + 1 = (5 \times 2^x + 1)2$

$\Rightarrow \dfrac{2}{2^x} + 1 = 10 \times 2^x + 2$

Let $2^x = y$

So, $\dfrac{2}{y} + 1 = 10 \times y + 2$

$2 + y = 10y^2 + 2y$

$\Rightarrow y = \dfrac{2}{5}$ or $\dfrac{-1}{2}$

i.e. $2^x = \dfrac{2}{5}$ $\left(\text{as } 2^x \neq -\dfrac{1}{2}\right)$

$x = 1 - \log_2 5$.

30. 2 Given :- $|x^2 - x - 6| = x + 2$

$\Rightarrow |(x - 3)(x + 2)| = x + 2$

For $x < -2$, $(3 - x)(-x - 2) = x + 2$

$\Rightarrow x - 3 = 1 \Rightarrow x = 4$ (Not possible, as $x < -2$)

For $-2 \le x < 3$, $(3 - x)(x + 2) = (x + 2)$

$\Rightarrow x = 2, -2$

For $x \ge 3$, $(x - 3)(x + 2) = x + 2$

$\Rightarrow x = 4$

Hence the product $= 4 \times 2 \times (-2) = -16$

31. 4 $\log_3 x = a \Rightarrow x = 3^a$

and $\log_{12} y = a \Rightarrow y = 12^a$

Now $G = \sqrt{xy} = \sqrt{3^a \times 3^a \times 4^a} = \sqrt{3^{2a} \times 2^{2a}} = 3^a \times 2^a$

$= 6^a$

Hence, $\log_6 G = \log_6 6^a = a$

32. $\dfrac{\log(2^3 \times 3^3 \times 5) + \log(2^6 \times 3 \times 5^7) + \log(2 \times 3^2 \times 5^4)}{3}$

$= \log(2^a \times 3^b \times 5^c)$

$\Rightarrow \log(2^8 \times 3^4 \times 5^8) + \log(2 \times 3^2 \times 5^4)$

$= 3\log(2^a \times 3^b \times 5^c)$

$\log(2^9 \times 3^6 \times 5^{12}) = 3\log(2^a \times 3^b \times 5^c)$.

$\Rightarrow 2^9 \times 3^6 \times 5^{12} = (2^a \times 3^b \times 5^c)^3$

$\Rightarrow 3a = 9, a = 3$.

33. 4 $2^x = 3^{\log_5 2}$

Taking log both sides

$\log 2^x = \log 3^{\log_5 2}$

$x \log 2 = \log_5 2 \times \log 3$

$x \log 2 = \dfrac{\log 2}{\log 5} \times \log 3$

$x = \dfrac{\log 3}{\log 5} + 1 - 1$

$x = \dfrac{\log 3 - \log 5}{\log 5} + 1$

$= \dfrac{\log \frac{3}{5}}{\log 5} + 1 = 1 + \log_5 \dfrac{3}{5}$

34. 3 $\text{Log}_5(x + y) + \log_5(x - y) = 3$

$\Rightarrow \log_5(x + y)(x - y) = 3$.

$\Rightarrow \log(x^2 - y^2) = 3\log 5$

$\Rightarrow x^2 - y^2 = 5^3$ ...(1)

$\log_2 y - \log_2 x = 1 - \log_2 3$

$\Rightarrow \log_2 \dfrac{y}{x} = \log_2 2 - \log_2 3$

$\Rightarrow \log_2 \dfrac{y}{x} = \log_2 \dfrac{2}{3}$

$\Rightarrow \dfrac{y}{x} = \dfrac{2}{3}$ ...(2)

Substitute the value of y from (2) in (1) and get,

$\Rightarrow x^2 - \left(\dfrac{2x}{3}\right)^2 = 5^3 \Rightarrow x^2 \dfrac{5}{9} = 5^3$

$\Rightarrow x = 15$ and $y = 10$

$\therefore xy = 150$.

35. 1 $2^{y^2 \log_3 5} = 5^{\log_2 3}$

$\Rightarrow y^2 \log_3 5 \log 2 = \log_2 3 \log 5$

$\Rightarrow y^2 = \dfrac{\log_2 3 \times \log 5}{\log 2 \times \log_3 5}$

$\Rightarrow y^2 = \left(\dfrac{\log 3}{\log 2}\right)^2$

$\Rightarrow y = \pm \dfrac{\log 3}{\log 2}$

Since $y < 0$, therefore, $y = -\dfrac{\log 3}{\log 2} = \log_2\left(\dfrac{1}{3}\right)$